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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	ERROR 303
Selected Domain	Domain 2
Selected SDG Track (1–17)	8,9,12
Project / Solution Name	StockWise
Name (Leader)	Timothy Lai Yong Fong
Email & Phone (Leader)	timothyfy851@gmail.com / +60 111325 8211
University Affiliation(s)	Taylor's University
Number of Team Members	3

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Nowadays, Malaysian SME still rely on manual handbooks, scattered Whatsapp groups or Excel spreadsheets to manage raw materials inventory. Managing these manually while responding to changing customer demand, unstable material prices and supply disruptions can be tedious and extremely inefficient.

This problem affects manufacturers, inventory managers, procurement teams, suppliers, and ultimately, customers. Small and medium-sized manufacturers may be particularly vulnerable because they often have fewer resources to monitor market trends, forecast demand, and manage unexpected price increases. Economically, material shortages and sudden price increases can reduce profitability, while overstocking increases operating expenses. Environmentally, inaccurate demand forecasting can contribute to unnecessary production, excess packaging, material waste, and avoidable transportation. Socially, production disruptions can affect employees' workloads, job stability, supplier relationships, and the timely delivery of essential products to customers.

Our team aims to address these challenges by developing an AI-powered inventory and predictive restocking system. The system will calculate raw material requirements from incoming customer orders, monitor inventory levels, predict potential shortages, and analyze relevant market news to identify possible price increases or supply disruptions. Based on these insights, it will recommend suitable restocking quantities and purchasing times.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills

gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

StockWise AI is an AI-assisted inventory management and procurement application designed to help businesses maintain optimal stock levels, reduce inventory waste, and make informed purchasing decisions. Unlike traditional inventory systems that primarily record stock quantities and generate low-stock alerts, StockWise AI aims to connect incoming orders, material requirements, current inventory, supplier information, and market trends to support proactive restocking decisions.

When a new order is received, the system analyses the materials required to fulfil the order and compares the estimated requirements with available inventory. It identifies potential shortages, estimates replenishment needs, and recommends appropriate purchasing quantities and timing. The proposed AI capabilities will also analyse historical consumption patterns, supplier lead times, and relevant market information to identify potential price fluctuations or supply risks, helping businesses decide whether to restock immediately or purchase additional materials in advance.

The application is designed for cross-industry use, including manufacturing, retail, restaurants, construction, healthcare, and educational institutions. Its recommendations can be adapted to different inventory types, operational requirements, and purchasing workflows.

Our team intends to leverage skills gained from the SDC Phase 1 workshops, including generative AI integration through APIs, database architecture, web application development, and cloud deployment. Generative AI can help interpret operational data and explain procurement recommendations, while structured databases will support inventory, order, and supplier records. Cloud hosting will enable the application to be accessed and tested through a deployed web interface.

The initial prototype establishes the core user interface and inventory management workflow. Phase 2 will focus on integrating AI services, persistent data storage, and reliable deployment to develop a more functional and testable intelligent restocking solution.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Component	Proposed Technology	Purpose
Frontend	HTML	Provides an interactive dashboard for managing inventory, orders, suppliers, and restocking recommendations.

Use of official SDC sponsor-provided GenAI API or Cloud Platform

The phase-1 prototype has used Replit to construct and design the layout.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We adopted a user-centred, iterative design-thinking framework to build an inventory web application addressing shortages, overstocking, and inefficient restocking:

- **Empathise:** Identified key operational challenges in stock tracking, material forecasting, and supplier coordination.
- **Define:** Identified the root problem as fragmented data such as orders, existing stock, suppliers, and market conditions obstructing timely, data-driven purchasing decisions.
- **Ideate:** Designed an AI-assisted engine evaluating customer orders, existing stock, lead times, and market trends to guide restocking.
- **Prototype:** Built an interactive HTML web prototype demonstrating core workflows through browser-based storage and rule-based logic.

- **Test:** Evaluated the prototype using realistic inventory scenarios, assessed calculation accuracy, shortage alerts, and usability across realistic operational scenarios.

Phase 2 Deployment Strategy Our team will transition to a production cloud architecture by integrating a persistent database, backend APIs, and GenAI-powered demand forecasting. Deliverables include a deployed web application, automated restocking alerts, and a testing report.

Deterministic calculations will ensure accurate inventory quantities, while AI will support analysis and explanations. Purchasing decisions will remain subject to human approval. This phased approach will help us deliver a reliable, user-friendly, and scalable solution suitable for multiple industries.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

*The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess **Innovation and Creativity** as well as **Feasibility of build**. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 8.4 "Achieve higher levels of economic productivity through technological upgrading and innovation." Target 12.2 "Achieve the sustainable management and efficient use of natural resources."
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Manual inventory replenishment recorded a 3.72% stockout rate in a study of 85 retail stores, demonstrating the real-world problem of inventory shortages. (Source: Sustainability, 2021).
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage reduction in inventory stockout rate (%) compared with the baseline.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	20% relative reduction in inventory stockout rate compared with the baseline during simulated prototype testing. If the baseline is 25%, the target is to reduce it to 20%.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.

- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

- ☐ Yes
- ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

- ☐ Yes
- ☑ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

- ☐ Yes
- ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



TIMOTHY LAI YONG FONG

24/10/2026

8. Reference List

Avlijas, G., Vukanovic Dumanovic, V., & Radunovic, M. (2021). Measuring the Effects of Automatic Replenishment on Product Availability in Retail Stores. Sustainability, 13(3), 1391. https://doi.org/10.3390/su13031391